Bylaw 2023

Fall 2025

Motor Assignment

Phase 3



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1. Develop a MATLAB script to determine optimal controller parameters Write a script that calculates the optimal values of Kp and Td that minimize the following formula.

Editor - D:\reports\control systems\motor assignment\optimize.m

first_order_parm.m | motor_initialization.m | second_order_param.m | iae.m | optimize.m | figure7.m | best_case_c

```
32
33              assignin('base', 'Kp', current_Kp);
34              assignin('base', 'Td', current_Td);
35
36              % Run Simulation
37              try
38
39                  res = sim(model_name);
40              catch ME
41                  warning('Sim failed for Kp=%.2f, Td=%.2f: %s', current_Kp, current_Td, ME.message);
42                  continue;
43              end
44
45
46              [e_vals, dt] = get_signal_data(res, 'simout2'); % Error signal
47              [u_vals, ~]  = get_signal_data(res, 'simout1'); % Control signal
48
49              % Calculate Cost
50              n = min(length(e_vals), length(u_vals));
51              e_vals = e_vals(1:n);
52              u_vals = u_vals(1:n);
53
54              if length(dt) > 1
55                  dt = dt(1:n);
56              else
57                  dt = repmat(dt, n, 1);
58              end
59
60              sum_error  = sum(abs(e_vals) .* dt);
61              sum_effort = sum(abs(u_vals) .* dt);
62
63              current_J = sum_error + (0.5 * sum_effort);
64
```

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```
1
2   function optimize_controller_robust
3       clc; clear; close all;
4       R=pi/2;
5       % Define Optimization Parameters
6       Kp_range = 5:5:30;        % Range for Kp
7       Td_range = 0.01:0.05:0.3; % Range for Td
8
9       model_name = 'figure4_motor';
10
11      if bdIsLoaded(model_name)
12          save_system(model_name);
13      else
14          load_system(model_name);
15          save_system(model_name);
16      end
17
18      min_J = inf;
19      best_Kp = 0;
20      best_Td = 0;
21
22      fprintf('Starting Optimization (Robust Mode)...\n');
23
24      %  Cascaded Loops
25      for i = 1:length(Kp_range)
26          for j = 1:length(Td_range)
27
28
29              current_Kp = Kp_range(i);
30              current_Td = Td_range(j);
```

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```
            if current_J < min_J
                min_J = current_J;
                best_Kp = current_Kp;
                best_Td = current_Td;
            end
        end
    end

    % Final Verification
    fprintf('\n--------------------------------------\n');
    fprintf('OPTIMIZATION COMPLETE\n');
    fprintf('Best Cost (J) : %.4f\n', min_J);
    fprintf('Best Kp       : %.2f\n', best_Kp);
    fprintf('Best Td       : %.3f\n', best_Td);
    fprintf('--------------------------------------\n');

    assignin('base', 'Kp', best_Kp);
    assignin('base', 'Td', best_Td);

    fprintf('Running final verification simulation...\n');
    res_final = sim(model_name);

    % for plotting
    [y_final, t_final] = get_signal_data(res_final, 'simout');   % Angle
    [u_final, ~]       = get_signal_data(res_final, 'simout1');  % Voltage

    figure('Name', 'Optimized Response');
    subplot(2,1,1);
    plot(t_final, y_final, 'LineWidth', 2);
    grid on; title(['Optimized Output (Kp=' num2str(best_Kp) ', Td=' num2str(best_Td) ')']);
    ylabel('Angle (rad)');
```

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```
    subplot(2,1,2);
    plot(t_final, u_final, 'r', 'LineWidth', 1.5);
    grid on; title('Control Effort'); ylabel('Volts'); xlabel('Time (s)');
end

% HANDLE ALL DATA TYPES
function [vals, time_vec] = get_signal_data(sim_res, var_name)
    % 1. Check if the variable exists in the SimulationOutput
    if ~isprop(sim_res, var_name)
        if isprop(sim_res, 'yout') && isa(sim_res.yout, 'Simulink.SimulationData.Dataset')
            try
                ds = sim_res.yout.get(var_name);
                vals = ds.Values.Data;
                time_vec = ds.Values.Time;
                return;
            catch
            end
        end
        error('Variable "%s" not found in simulation output.', var_name);
    end

    raw_data = sim_res.(var_name);

    %  "Timeseries" Object
    if isa(raw_data, 'timeseries')
        vals = raw_data.Data;
        time_vec = raw_data.Time;

    %Structure with Time"
    elseif isstruct(raw_data) && isfield(raw_data, 'signals') && isfield(raw_data, 'time')
        vals = raw_data.signals.values;
        time_vec = raw_data.time;
```

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```
129
130         % "Array"
131         elseif isnumeric(raw_data)
132             if isprop(sim_res, 'tout')
133                 time_vec = sim_res.tout;
134                 if length(raw_data) == length(time_vec)
135                     vals = raw_data;
136                 else
137                     [r, c] = size(raw_data);
138                     if c > 1 && length(raw_data) == length(time_vec)
139                         vals = raw_data(:, 2:end);
140                     else
141                         vals = raw_data;
142                     end
143                 end
144             else
145                 vals = raw_data;
146                 time_vec = 1:length(vals);
147             end
148
149         % "Dataset" Object
150         elseif isa(raw_data, 'Simulink.SimulationData.Dataset')
151              el = raw_data.get(1);
152              vals = el.Values.Data;
153              time_vec = el.Values.Time;
154         else
155             error('Unknown data format for variable %s.', var_name);
156         end
157     end
```

OPTIMIZATION COMPLETE

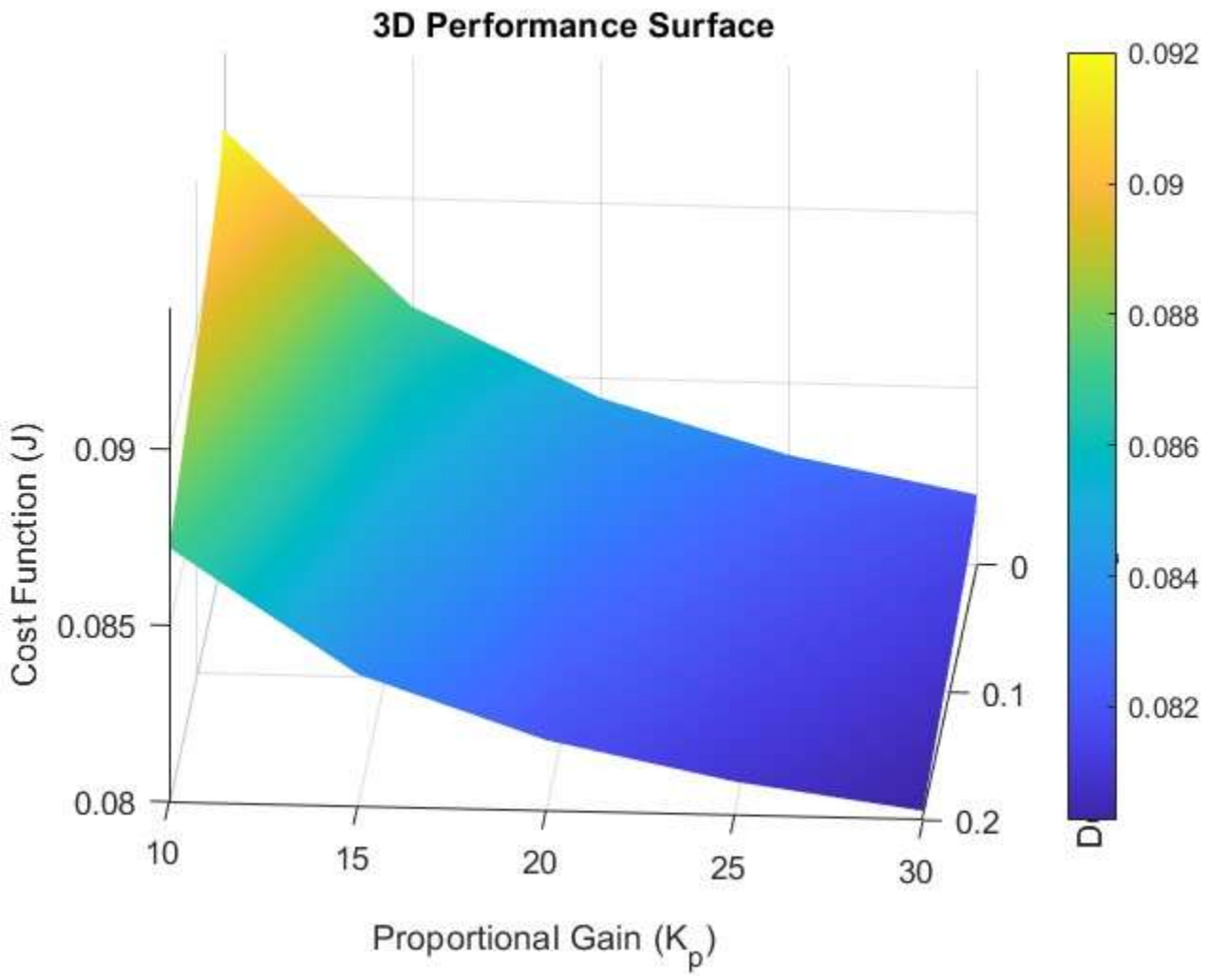
Best Cost (J) : 9.2327

Best Kp : 5.00

Best Td : 0.010

2. Plot the 3D performance surface

Generate a 3D surface plot showing the relationship between Kp, Td, and IAE.



3. Simulate the system using the optimal controller
   Using the best (optimal) Kp and Td, simulate the system response for a square-wave input with:
   - Frequency = $\frac{0.1}{T_m}$
   - Amplitude alternating between $\pm\frac{\pi}{2}$

